

DYNAMICS OF TILING BILLIARDS ON APERIODIC HAT TILINGS

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ABSTRACT. We study tiling billiards on hierarchical aperiodic tilings by the hat monotile. Using computational experiments, we observe substantial geometric organization in trajectories, including three direction-dependent visual styles, recurring local motifs, spatial clustering, extensive path reversal, and abundant seemingly periodic behavior. In particular, we conjecture that every nonsingular trajectory in one distinguished class of hat-edge directions is periodic. The apparent prevalence of periodicity parallels preliminary observations previously reported for Penrose tiling billiards and suggests that strong recurrence may be a broader feature of billiards on aperiodic tilings. We also study the symbolic dynamics obtained by recording successive hat crossings. We prove that the global language complexity of nonsingular trajectories grows at most quintically, and hence that the associated global symbolic system has topological entropy zero. We conjecture that this bound can be substantially improved and that the language complexity of every individual nonsingular trajectory grows linearly.

1. INTRODUCTION

Tiling billiards are piecewise-linear dynamical systems on planar tilings. A trajectory travels in a straight line within each tile and, upon crossing a tile edge, refracts according to the negative-refraction rule with coefficient -1 . Equivalently, the trajectory's outgoing direction after crossing an edge is the reflection of its incoming direction across said edge. Trajectories meeting a vertex of a tile are deemed singular and terminate as their continuation is not uniquely determined. Tiling billiards were introduced as a mathematical dynamical system by Diana Davis, Kelsey DiPietro, Jenny Rustad, and Alexander St Laurent, in work arising from a problem introduced by Sergei Tabachnikov and motivated by earlier work on negative refraction [4, 5].

Although tiling billiards can be defined on arbitrary planar tilings, the published dynamical theory has so far focused on periodic tilings. These include periodic triangle tilings [2, 6], periodic cyclic-quadrilateral tilings [8], and the trihexagonal tiling [5]. See [7, §1.1] for a recent bibliographical overview of the subject. Periodicity of the tiling means the local geometry of a tiling will repeat globally under translation. Consequently, the behavior of a trajectory can be captured in a finite fundamental domain, making tools such as foldings, interval exchange transformations with flips, and return maps common tools [8, 7].

Aperiodic tilings on the other hand do not have this local-to-global property, and so the aforementioned methods, which reduce the dynamics to a repeated finite region, are much more difficult to use. Predicting the long-term behavior of a trajectory instead requires some means of understanding how its local environment is situated within the tiling as a whole.

Tiling billiards on aperiodic tilings had nevertheless been proposed as a direction for further study in the original work of Davis et al., which specifically suggested the Penrose tiling [4]. Davis later reported preliminary joint investigations with Hooper on Penrose tiling billiards, with their experiments suggesting that almost every trajectory may be periodic [3].

In this paper, we study tiling billiards on the aperiodic tilings defined by the hat polykite. The hat's discovery answered the long-awaited question of whether or not a monotile could admit an aperiodic tiling—it was found by Smith, Myers, Kaplan, and Goodman Strauss [9].

They describe the tiling’s local-to-global organization through a substitution system on four metatiles H , T , P , and F . The substitution hierarchy decomposes a hat tiling into metatiles and then into successively larger supertiles, providing geometric structure at arbitrarily large scales. Our approach replaces translational periodicity with this hierarchy: rather than reducing the dynamics to a single repeating fundamental region, we study how trajectories interact with nested supertile structure as the relevant scale increases. These tilings are a strong choice to begin studying tiling billiards on aperiodic tilings as they contain significant order compared to other aperiodic tilings; this helps us study aperiodicity’s effect on the tiling billiards system without needing a plethora of new tools.

It is not currently known whether every tiling by the hat admits the standard global HTPF substitution hierarchy; see [9, p. 48]. We therefore restrict our attention to hat tilings that do admit such a hierarchy.

This paper investigates tiling billiards on hierarchical hat tilings. We study this system from both computational and theoretical perspectives. Using a simulator built on Kaplan’s *hatviz*, we find strong recurring structure in hat-tiling billiard trajectories, including three persistent visual styles, extensive path reversal, recurring local motifs, and abundant seemingly periodic behavior, similar to Davis’ and Hooper’s investigation of tiling billiards on the aperiodic Penrose tiling [3]. These experiments also suggest connections between trajectory behavior and the HTPF hierarchy and motivate several conjectures concerning periodicity. On the theoretical side, we impose a symbolic coding of tiling billiard trajectories. Although the aperiodicity of the tiling permits distinct geometric contexts at arbitrarily large scales, we prove that the number of distinct symbolic factors of length n encountered along all nonsingular trajectories grows at most quintically in n . The proof uses the substitution hierarchy of the hat to control how new trajectory behavior can arise across scales. A corollary is that the global tiling billiards system on hat tilings has topological entropy zero. We conjecture that individual trajectories have language complexity growing linearly in n . Together, these results provide an initial picture of how aperiodicity and substitution structure influence tiling billiards and suggest directions for their further study.

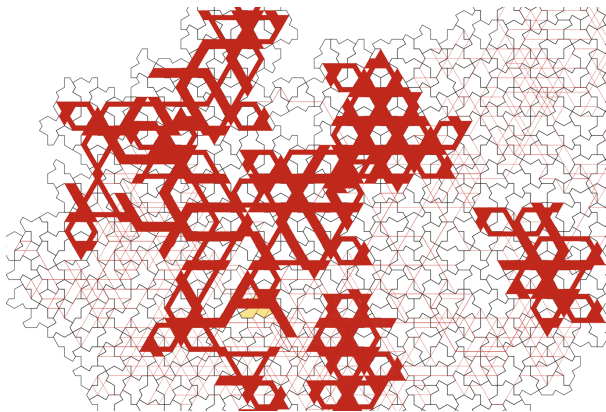


FIGURE 1. An example of a simulated tiling billiard trajectory with 114062 bounces on a hat polykite tiling. The thick red lines indicate places where the trajectory retraces itself very nearby whereas the fine red lines indicate places where this does not happen.

2. BACKGROUND AND DEFINITIONS

In this section we state known features of tilings by the hat and general definitions necessary for our investigation.

2.1. Tiling billiards. Tiling billiards are dynamical systems on planar tilings in which a trajectory travels linearly within each tile and refracts across tile boundaries with coefficient -1 . Equivalently, when a trajectory crosses an edge, its direction is reflected across the line containing that edge [4, Section 1.2]. Here we define tiling billiard trajectories and discuss a common technique called folding.

Definition 2.1 (Tiling billiard trajectory). Let τ be a tiling of the plane. A *tiling billiard trajectory* γ is a piecewise-linear path that travels in a straight line within each tile and, upon crossing an edge into an adjacent tile, refracts with coefficient -1 . Equivalently, the outgoing direction is obtained by reflecting the incoming direction across the crossed edge.

A trajectory that passes through a vertex of the tiling is called *singular*, since its continuation may not be uniquely defined. Throughout this paper, we refer to a tiling billiard trajectory simply as a *trajectory*, and we write Γ_τ for the set of all trajectories on τ (this includes both singular and nonsingular trajectories).

Trajectories in this paper are one-sided: each trajectory has a specified initial condition and evolves forward in time. We only allow initial points of trajectories to lie on an edge of a hat in τ .

Remark 2.2. Tiling billiard trajectories are sometimes taken to be bi-infinite, with the dynamics defined in both forward and backward time. A bi-infinite nonsingular trajectory may be viewed as the concatenation of two compatible one-sided trajectories satisfying the negative-refraction rule at their common initial point.

Because the outgoing direction of a trajectory after crossing an edge is the reflection of the incoming direction across that edge, we may reflect the newly entered tile across the crossed edge. Under this folding, the incoming and outgoing trajectory segments lie on the same straight line. Repeating this process along a trajectory therefore folds all of its successive segments onto a single straight line [8, Section 2.1] while all the hats crossed by the trajectory now overlap. The folding technique is common in tiling billiards.

2.2. The hat and the HTPF hierarchy. There are several geometric features of the hat and its tilings that we will use throughout this paper. The hat is a 13-sided polykite formed from eight kites in the [3.4.6.4] Laves tiling [9, Section 1.2]; we enumerate its sides according to Figure 2a. Moreover, every tiling by the hat is aligned with an underlying [3.4.6.4] Laves tiling [9, Lemma A.6]. Thus the constituent kites of all hats in the tiling fit together into a single common kite grid, as illustrated in Figure 2b.

Remark 2.3. Hat tilings contain both the hat and its reflected copy. We use the same side-labeling convention for both orientations, with corresponding sides receiving the same labels.

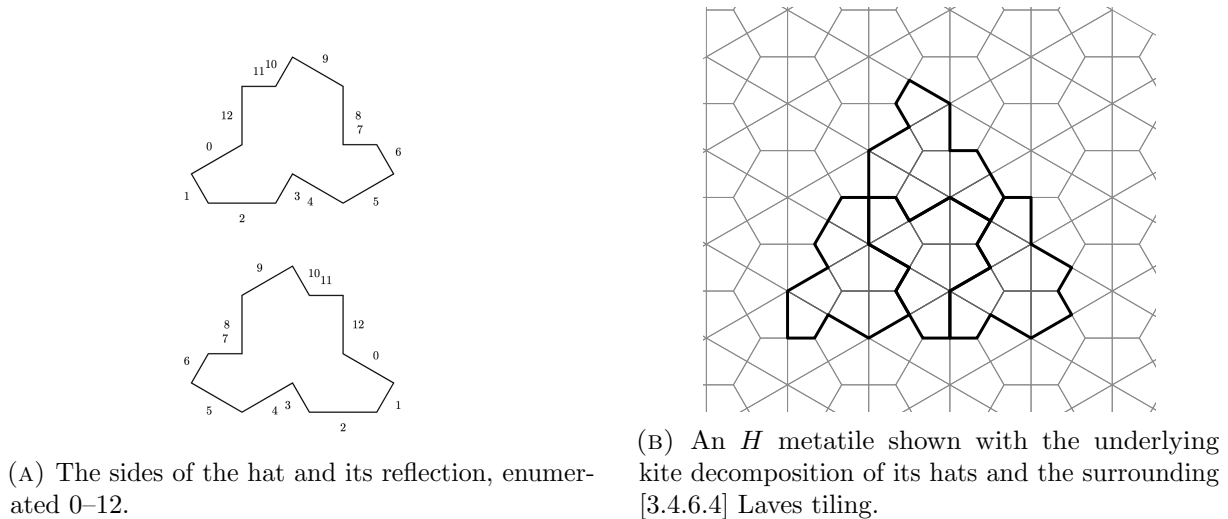


FIGURE 2. The hat polykite, its side labeling, and its underlying kite structure.

Hat tilings have hierarchical structure beyond the underlying kite grid. In the HTPF construction, clusters of hats are grouped into four metatyle types, denoted H , T , P , and F . These metatiles fit together to form larger, combinatorially equivalent supertiles with the same four types and matching structure, allowing the construction to be iterated recursively [9, Theorem 5.1]. The associated substitution has linear inflation factor ϕ^2 , where $\phi = (1 + \sqrt{5})/2$ is the golden ratio, and hence area growth factor ϕ^4 [9, Section 2]. Smith et al. additionally report computationally that, after normalization for size, the four sequences of supertile shapes converge to four limiting shapes [9, Section 2]. We refer to this system as the *HTPF hierarchy* for the remainder of this paper. Below we define the class of hierarchical hat tilings considered in this paper.

Definition 2.4 (Hat tiling). A *hat tiling* is a tiling of the plane by copies of the hat polykite and its reflection that admits a global HTPF substitution hierarchy as described in *An Aperiodic Monotile* [9]. Unless stated otherwise, every hat tiling considered in this paper is assumed to admit such a hierarchy.

Remark 2.5. SMKGS note the possibility that infinite hierarchical sectors could be joined along fault lines lying on supertile boundaries at every level [9, Section 7]. Such tilings, if they exist, need not be represented by a single global HTPF hierarchy and are therefore excluded from Definition 2.4.

2.3. Patches and hierarchical crossings. We now introduce notation for describing the interaction between tiling billiard trajectories and the HTPF hierarchy.

Definition 2.6 (Hat adjacency graph). Let τ be a hat tiling. The *hat adjacency graph* of τ , denoted by G_τ , is the simple undirected graph whose vertices correspond to the individual hats of τ . Two distinct vertices are joined by an edge if and only if the boundaries of the corresponding hats intersect in a line segment of positive length. Hats that meet only at isolated points are not adjacent in G_τ .

We write d_{G_τ} for graph distance in G_τ .

Definition 2.7 (Hat patch). A *hat patch* P in a hat tiling τ is a connected collection of hats, equivalently a connected subgraph of the hat adjacency graph G_τ .

The *geometric realization* of P , denoted by $|P|$, is the union of the closed subsets of the plane occupied by the hats in P . The *boundary* of P is the Euclidean boundary

$$\partial P := \partial|P|.$$

Remark 2.8. In some contexts, the term *patch* is reserved for a graph ball. We do not impose that restriction here: a hat patch may be any connected subgraph of G_τ .

Definition 2.9 (Level- k supertile). A level-0 supertile is a single hat, and a level-1 supertile is one of the four metatiles H , T , P , or F .

For each $k \geq 1$, the level- $(k+1)$ supertiles are obtained from the level- k supertiles by one application of the HTPF supertile construction. Every level- k supertile with $k \geq 1$ therefore has one of the four types H , T , P , or F .

The recursive HTPF construction is described in detail in [9, Section 5].

Definition 2.10 (Level- k seam). Let S_1 and S_2 be neighboring level- k supertiles in a hat tiling, and let P_1 and P_2 denote their associated hat patches. The *level- k seam* between S_1 and S_2 is the union of all positive-length line segments contained in

$$\partial P_1 \cap \partial P_2.$$

Definition 2.11 (Level- k crossing). A *level- k crossing* is an event in which a trajectory passes through a level- k seam from one level- k supertile into a neighboring level- k supertile in a hat tiling. A level-0 crossing is called a *bounce*.

3. OBSERVATIONS AND EXPERIMENTS

We began our study by creating a simulation of tiling billiards on hat tilings. Our experiments and observations from this simulation unveiled a number of interesting phenomena regarding this system. We conjecture several properties of this system as a result of our experiments and discuss unusual billiard behavior specific to hat tilings. The claims in this section are exploratory and should not be treated as rigorous results.

3.1. Methodology. To begin studying this system, we designed a simulation to run experiments within.

We implemented a tiling-billiards simulator on supertile patches generated using Craig Kaplan's *hatviz*. All observations in this section were obtained from this simulation. The source code is available in the author's [GitHub](#) repository.

The implementation uses floating-point arithmetic. In particular, coordinates involving $\sqrt{3}$ are represented numerically rather than exactly. Small errors may therefore accumulate during the construction of large patches and over long trajectory computations. Unless explicitly verified by exact calculation, the phenomena described in this section should be interpreted as computational observations rather than rigorous results.

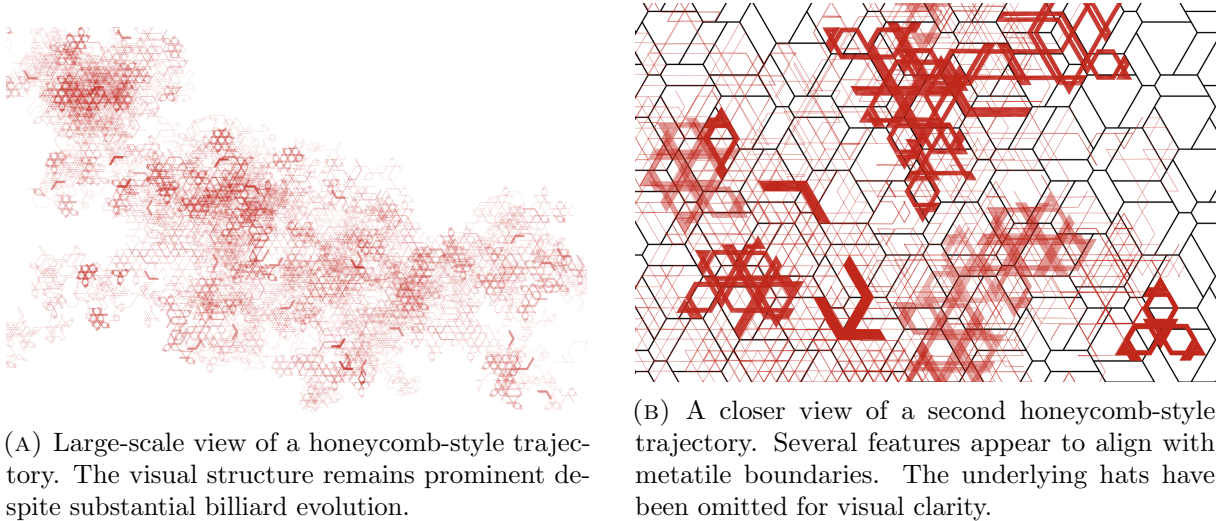
Our experiments consisted of two distinct approaches. The first is visual surveys: We manually ran trajectories and noted their features visually. Overall, we visually examined hundreds of trajectories and any results based on visuals were obtained from this sample. The lengths of these visuals were typically between 10000 and 100000 bounces. Smaller experiments were also performed in some cases, typically containing only tens or hundreds of bounces. Most of our surveyed trajectories began at a tile close to the center of a level-6 H supertile, though some had varied starting tiles. This center tile is the default in our simulation.

The second is batch runs: simulating thousands of trajectories with varied initial conditions but not examining them visually. Individual numbers were pulled from these trajectories and statistically analyzed. The bounce count and initial conditions were specific to individual experiments.

All trajectory runs were performed inside of a level-6 H supertile.

3.2. Three observed trajectory styles. In our exploratory surveys, trajectories became visually classifiable into three distinct styles after a short initial segment. We call these the *honeycomb*, *hexagonal*, and *thicket* styles. These terms describe qualitative visual patterns rather than formally defined dynamical classes. Each style contains recurring geometric features, examples of which are presented below.

We begin with the honeycomb style, examples of which are shown in Figure 3.



(A) Large-scale view of a honeycomb-style trajectory. The visual structure remains prominent despite substantial billiard evolution.

(B) A closer view of a second honeycomb-style trajectory. Several features appear to align with metatyle boundaries. The underlying hats have been omitted for visual clarity.

FIGURE 3. Examples of the honeycomb trajectory style. Recurring features include the Sigil and the V, which are discussed in Figure 5 and Figure 7 respectively.

The honeycomb style is characterized by a lattice-like structure resembling a honeycomb. It appears to contain two components: an underlying web of relatively sparse trajectory segments and a collection of recurring local patterns produced by extensive retracing. Both are visible in Figure 3b, where the thin lines form the web and the darker, thicker-looking regions result from the trajectory repeatedly following nearly the same path.

In many examples, a trajectory follows the web until it enters a small configuration of hats in which the local geometry causes it to retrace a similar path for a large number of bounces. These configurations produce the visually prominent recurring patterns described below. We have not attempted a complete classification of such patterns, but several examples are presented.

In our experiments, each recurring pattern appeared to be associated with a consistent collection of metatiles. In the following figures, we identify these collections as the metatiles through which the pattern visually travels and refer to it informally as the pattern's *metatile support*.

Some trajectory segments may briefly extend outside the hat patch associated with the support. For example, a trajectory can momentarily leave an H metatile despite every bounce being on its hat patch due to the concavity of the patch's corner. Nevertheless, every bounce of the recurring pattern interacts with the hats of the highlighted collection, so the pattern can be considered independently of the neighboring metatiles for the purposes of these illustrations. We do not claim that this visual support is uniquely determined.

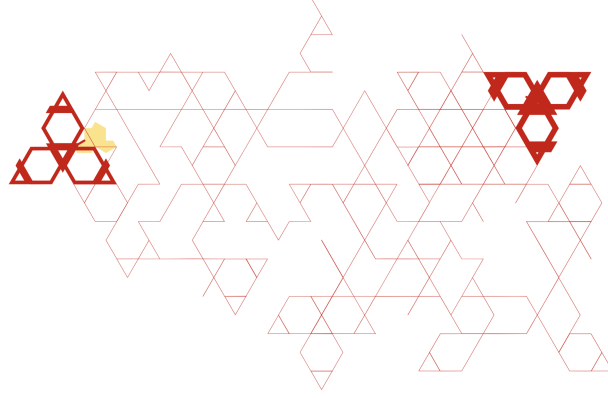
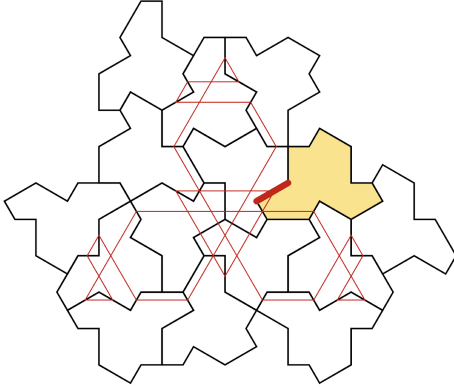
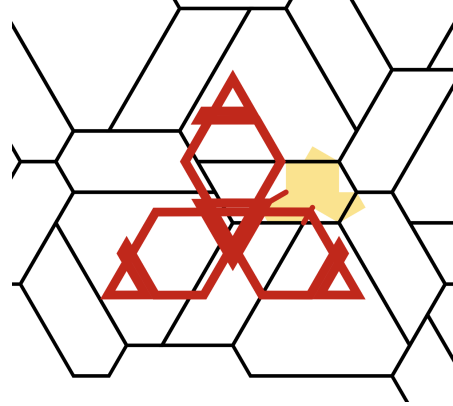


FIGURE 4. The web structure of a honeycomb-style trajectory linking two copies of the Sigil.

We make this clarification because metatiles visual boundaries can be replaced by arbitrary arcs for the purpose of generating a hat tiling, meaning a trajectory traveling inside the visual boundary of metatiles has no meaning in general.

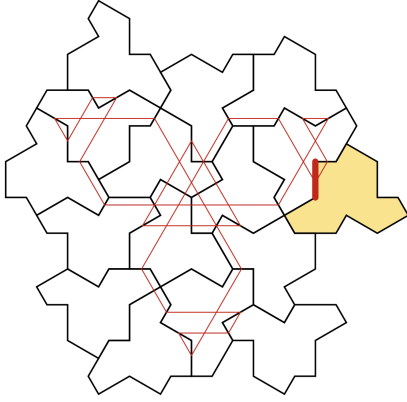


(A) A 36-bounce periodic trajectory exhibiting the Sigil. Exact bounce data verifying its periodicity are given in Table 2.

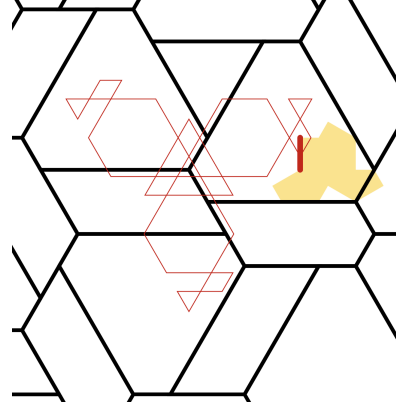


(B) A trajectory whose initial direction differs from that in (A) by 0.1° . After 1600 bounces, it remains localized along the Sigil pattern.

FIGURE 5. The Sigil, a recurring retracing pattern observed in honeycomb-style trajectories. Perturbed trajectories may repeatedly trace the Sigil before leaving the local hat configuration in which it occurs. In our experiments, the Sigil occurred only in the metatyle configuration shown in (B).

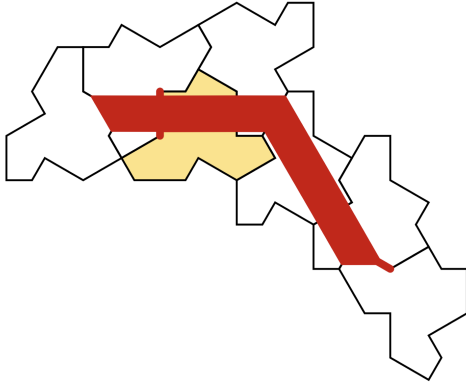


(A) The reflected Sigil and its local hat configuration.

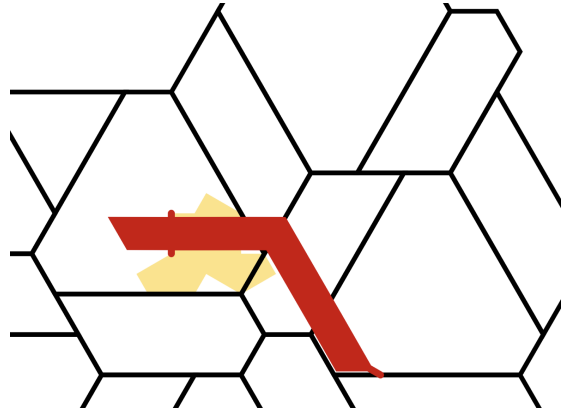


(B) The metatile arrangement surrounding the reflected Sigil shown in (A).

FIGURE 6. A reflected form of the Sigil also occurs in honeycomb-style trajectories. In our experiments, it appeared only in a metatile arrangement distinct from that of the unreflected Sigil.



(A) The V in its local hat configuration. This is the trajectory segment from Figure 20, continued for 1000 bounces rather than 12.

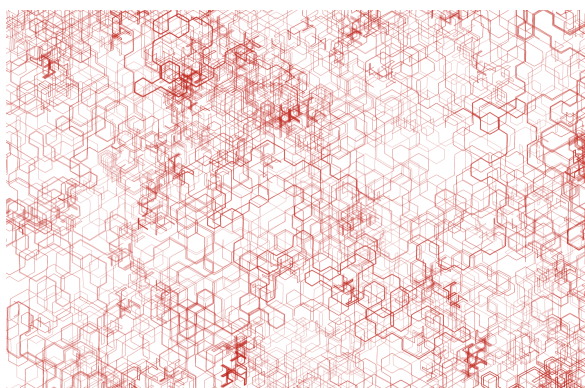


(B) The metatile arrangement surrounding the V shown in (A).

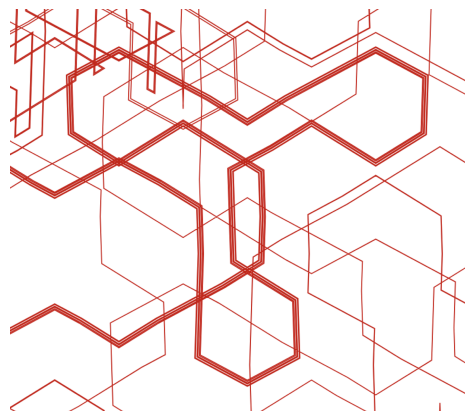
FIGURE 7. The V, a recurring retracing pattern observed in honeycomb-style trajectories.

We next consider the hexagonal style. As in the honeycomb case, hexagonal-style trajectories contain recurring local patterns, although these are often less visually prominent. One example is shown in Figure 8.

A second notable feature is the resemblance of some trajectory regions to other members of the hat family, particularly the comet tile. The frequent appearance of hexagonal structure is interesting in relation to the hexagonal representation of hat tilings described in [9, p. 11].

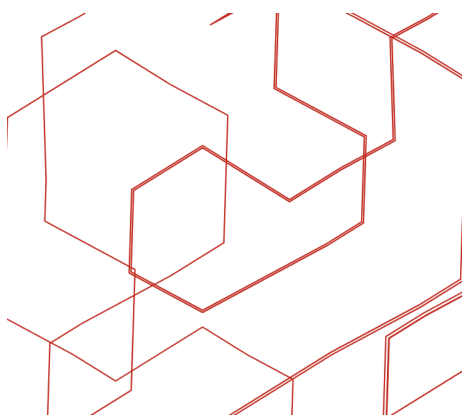


(A) A large-scale view of a hexagonal-style trajectory.

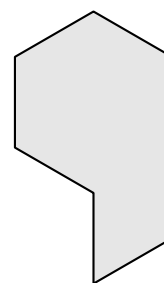


(B) A recurring local pattern observed in many hexagonal-style trajectories.

FIGURE 8. Examples of the hexagonal trajectory style and one of its recurring local patterns.



(A) A region of a hexagonal-style trajectory resembling the reflection of the comet member of the hat family.



(B) The comet tile $\text{Tile}(1,0)$.

FIGURE 9. Some regions of hexagonal-style trajectories visually resemble other members of the hat family. The example in (A) resembles the comet tile shown in (B).

Finally, we consider the thicket style. It is characterized by densely interwoven, tangled trajectory segments and a substantially wider range of visible segment directions than the honeycomb and hexagonal styles. Among the three observed styles, it appears the least geometrically regular. Like the other styles, it contains recurring local retracing patterns, two of which are shown below.

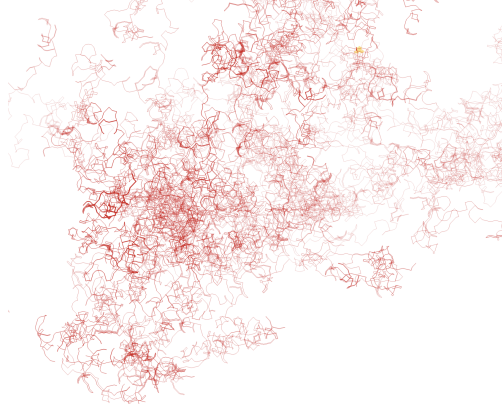
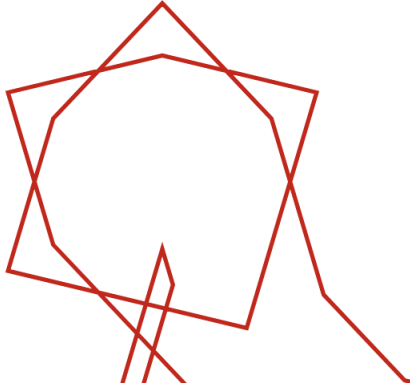


FIGURE 10. A long thicket-style trajectory. The twisted, intricate segments make this the least visually regular of the three observed styles.



(A) The Star, a recurring pattern observed in thicket-style trajectories.



(B) The Crescent, a recurring pattern observed in thicket-style trajectories.

FIGURE 11. Two recurring local patterns observed in thicket-style trajectories.

Across the starting positions tested in our experiments, trajectory style appeared to depend only on the initial direction. Because the edges of the hat in hat tilings occur in only finitely many directions due to the underlying kite grid, reflection across hat edges restricts each trajectory to a finite set of segment directions determined by its initial angle. This directional constraint appears to produce the persistent large-scale visual structure associated with each style.

The approximate simulation angle ranges corresponding to the three observed styles are listed in Table 1. These ranges were determined visually from a 1° angle sweep covering 360 trajectories on the default simulation tile. We varied the starting side when trajectories seemed periodic in order to ensure the style holds over longer bounce counts. Despite this experiment being based on one tile, these buckets appear consistent on other starting tiles. Near the boundaries between these ranges, the characteristic structure of a style becomes less stable, and features associated with two different styles can occur in the same trajectory. We did not find this behavior away from these boundaries in our experiments. Examples are shown in Figure 12.

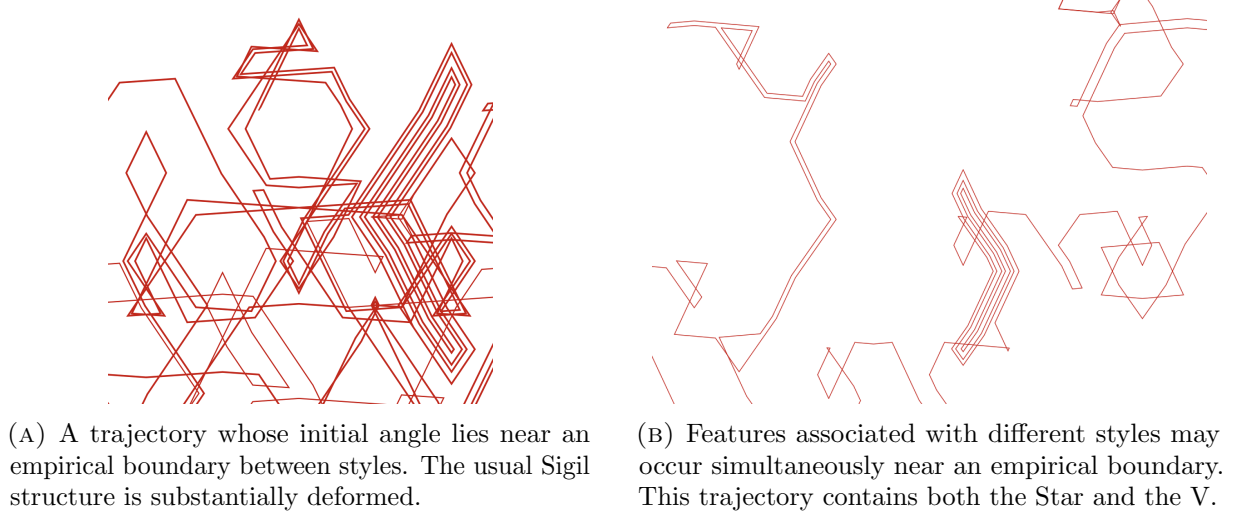


FIGURE 12. Trajectory behavior near empirical boundaries between the three observed visual styles.

TABLE 1. Trajectory style by approximate initial simulation angle

Angle interval	Trajectory style
$0^\circ-5^\circ$	Honeycomb
$5^\circ-24^\circ$	Thicket
$24^\circ-35^\circ$	Hexagonal
$35^\circ-56^\circ$	Thicket
$56^\circ-67^\circ$	Honeycomb
$67^\circ-84^\circ$	Thicket
$84^\circ-95^\circ$	Hexagonal
$95^\circ-115^\circ$	Thicket
$115^\circ-126^\circ$	Honeycomb
$126^\circ-144^\circ$	Thicket
$144^\circ-157^\circ$	Hexagonal
$157^\circ-174^\circ$	Thicket
$174^\circ-187^\circ$	Honeycomb
$187^\circ-204^\circ$	Thicket
$204^\circ-218^\circ$	Hexagonal
$218^\circ-235^\circ$	Thicket
$235^\circ-247^\circ$	Honeycomb
$247^\circ-263^\circ$	Thicket
$263^\circ-277^\circ$	Hexagonal
$277^\circ-295^\circ$	Thicket
$295^\circ-306^\circ$	Honeycomb
$306^\circ-325^\circ$	Thicket

Angle interval	Trajectory style
325°–339°	Hexagonal
339°–354°	Thicket
354°–360°	Honeycomb

The approximate pattern of angle ranges seems to repeat with a period of 60°, consistent with the finite set of edge directions occurring in the underlying tiling.

3.3. Periodicity and trajectory exploration. As discussed earlier, several recurring patterns cause trajectories to remain localized inside a small hat configuration while repeatedly retracing a pattern. This behavior is especially prominent in the honeycomb style and makes these patterns natural candidates for periodic orbits. We verified by exact calculation that the Sigil admits a periodic realization. Periodicity has not yet been tested systematically for the other recurring patterns.

TABLE 2. Exact bounce data for one period of the Sigil trajectory.

Bounce number	Coordinates	Angle after bounce (degrees)
0	$(\frac{1}{4}, \frac{3\sqrt{3}}{4})$	180
1	$(-2, \frac{3\sqrt{3}}{4})$	180
2	$(\frac{-13}{4}, \frac{3\sqrt{3}}{4})$	120
3	$(\frac{-9}{2}, 2\sqrt{3})$	60
4	$(-2, \frac{9\sqrt{3}}{2})$	300
5	$(\frac{-5}{4}, \frac{15\sqrt{3}}{4})$	180
6	$(\frac{-15}{4}, \frac{15\sqrt{3}}{4})$	240
7	$(\frac{-17}{4}, \frac{13\sqrt{3}}{4})$	0
8	$(\frac{-3}{4}, \frac{13\sqrt{3}}{4})$	300
9	$(\frac{1}{2}, 2\sqrt{3})$	240
10	$(\frac{-13}{8}, \frac{-\sqrt{3}}{8})$	240
11	$(\frac{-5}{2}, -\sqrt{3})$	300
12	$(-2, \frac{-3\sqrt{3}}{2})$	60
13	$(\frac{-7}{8}, \frac{-3\sqrt{3}}{8})$	60
14	$(\frac{-1}{4}, \frac{\sqrt{3}}{4})$	0
15	$(\frac{9}{4}, \frac{\sqrt{3}}{4})$	300
16	$(\frac{19}{4}, \frac{-9\sqrt{3}}{4})$	180
17	$(\frac{13}{4}, \frac{-9\sqrt{3}}{4})$	60
18	$(\frac{9}{2}, -\sqrt{3})$	120
19	$(4, \frac{-\sqrt{3}}{2})$	240
20	$(\frac{9}{4}, \frac{-9\sqrt{3}}{4})$	180
21	$(\frac{-1}{4}, \frac{-9\sqrt{3}}{4})$	120
22	$(\frac{-19}{8}, \frac{-\sqrt{3}}{8})$	120
23	$(\frac{-13}{4}, \frac{3\sqrt{3}}{4})$	180
24	$(\frac{-17}{4}, \frac{3\sqrt{3}}{4})$	300

Bounce number	Coordinates	Angle after bounce (degrees)
25	$(\frac{-25}{8}, \frac{-3\sqrt{3}}{8})$	300
26	$(\frac{-5}{2}, -\sqrt{3})$	240
27	$(\frac{-15}{4}, \frac{-9\sqrt{3}}{4})$	180
28	$(\frac{-25}{4}, \frac{-9\sqrt{3}}{4})$	120
29	$(-8, \frac{-\sqrt{3}}{2})$	240
30	$(\frac{-17}{2}, -\sqrt{3})$	300
31	$(\frac{-29}{4}, \frac{-9\sqrt{3}}{4})$	180
32	$(\frac{-35}{4}, \frac{-9\sqrt{3}}{4})$	60
33	$(\frac{-25}{4}, \frac{\sqrt{3}}{4})$	0
34	$(-2, \frac{\sqrt{3}}{4})$	0
35	$(\frac{-1}{4}, \frac{\sqrt{3}}{4})$	60
36	$(\frac{1}{4}, \frac{3\sqrt{3}}{4})$	180

The position and outgoing direction at bounce 36 agree with those at bounce 0. Since the trajectory returns to the same phase-space state for the first time at bounce 36, the orbit is periodic with minimal period 36.

Our experiments also suggest a distinguished collection of directions for which periodic behavior may be universal. This leads to the following conjecture:

Conjecture 3.1 (Edge-parallel periodicity). Fix a hat tiling τ . Every nonsingular trajectory $\gamma \in \Gamma_\tau$ that is at some point parallel to any edge of any hat in τ is periodic. Equivalently, any trajectory with an initial angle parallel to an edge of a hat in τ is periodic.

We conjecture this as we were unable to find trajectories that did not seem to be periodic along these directions, regardless of starting conditions. Of course, we did not verify that these trajectories were truly periodic due to the limitations of our simulation. Given the alignment of this angle class with the underlying kite grid of hat tilings, however, there is some geometric motivation to this claim.

We attempted to test this theory by measuring how many unique hats trajectories tend to visit based on their angle over many bounces. Let M be the number of unique hats a trajectory of n bounces visits divided by n . Periodic trajectories should loop over the same hats while accumulating bounces, making $M \rightarrow 0$ as $n \rightarrow \infty$. So to test Conjecture 3.1, we check whether M tends to 0 near multiples of 30° . Of course, this method is flawed for several reasons. Long periodic trajectories may exist, so they may never see a cycle in our chosen bounce count. Unique hats visited may also not constitute clear evidence for periodicity as hat trajectories have a significant amount of retracing and clustering, which is discussed in the next section. Some small angle deviations will also still loop around equivalent hat-patches for some time; the length of this experiment may not capture their eventual divergence. Still, the results show a clear trend: approaching the angles of the lattice direction pushes M down while moving away from these directions pushes it up. The experiment is described below.

We chose 3 random hats within our level-6 H supertile. Then on each hat, we randomly selected an edge and a coordinate on that edge for trajectories to start from. We proceeded to simulate 5000-bounce trajectories at 0.1° increments starting from 0° until the angle sweep returned to its start (3600 trajectories per hat). For each trajectory, we measured the number of unique hats the trajectory visited divided by its bounce count and recorded the measure's corresponding launch angle. By convention, a trajectory visits a hat if it intersects an edge of

said hat. As such, $0 \leq M \leq 2$. We handled trajectories that were either singular or left the supertile by omitting their M . In all 10800 trajectories, this only occurred 3 times so it will not meaningfully influence our calculations. One failure occurred in tile 1 (Figure 13a) and the others in tile 3 (Figure 13c). The results, along with starting conditions for each experiment, are shown in Figure 13.

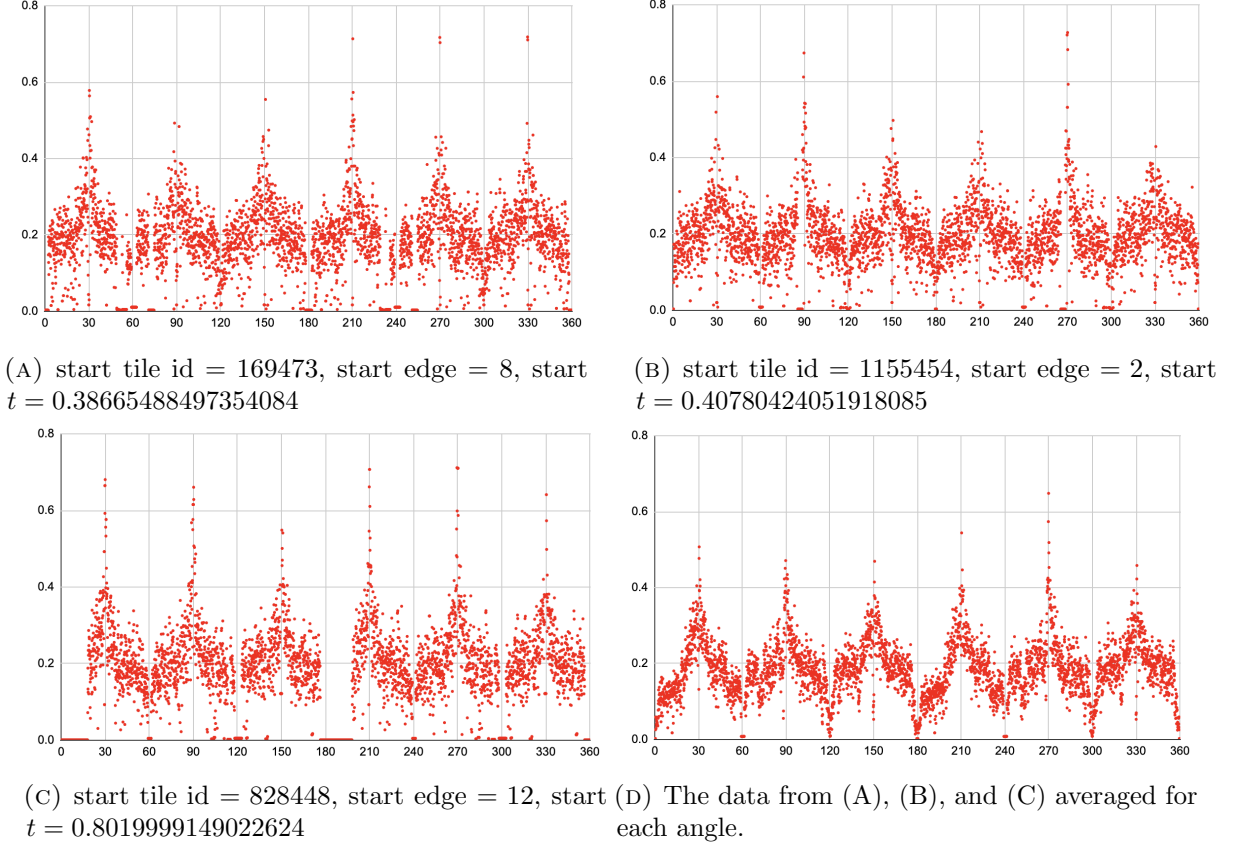


FIGURE 13. (A), (B), and (C) show launch angle on the x-axis and M on the y-axis for the angle's associated trajectory. (D) is the average of the M values at each angle for (A), (B), and (C).

The data set of M grouped across all 10797 remaining trajectories has the following metrics: $Max = 0.7276$, $mean = 0.1826$, $median = 0.1874$, $stdev = 0.1001$ all truncated to 4 decimal places. The mean and median are consistent with the individual trajectories we have observed. This data, along with the graphs in Figure 13, suggests a high amount of re-visitation among hat trajectories and supplies evidence for Conjecture 3.1 in half of the hat-edge directions, but seems to contradict the others. M reaches its minimum values near hat-edge directions of $(60n)^\circ$. Surprisingly, M appears to attain its maximum values near hat-edge directions of $(60n + 30)^\circ$. Both direction classes are geometrically distinguished by the edge structure of the tiling, yet they exhibit apparently opposite behavior in this experiment. We believe trajectories with initial angles of $(60n + 30)^\circ$ explore many more hats than trajectories of the other direction class before completing one period. This belief is based on our visually surveyed trajectories—many trajectories at $(60n + 30)^\circ$ required 20000–100000 bounces before our simulation detected a cycle. Cycles are not defined in the simulation as certificates of periodicity—they mark extremely close

returns to initial conditions. Our belief is also consistent with high values of M for trajectory bounce-counts near 5000. We know of no mechanism to explain this difference in exploration speed or apparent periodicity. We had difficulty in testing this hypothesis with increased bounce-counts as many trajectories left our chosen level-6 H supertile before completing the required number of bounces; we were unable to render larger supertiles with our equipment. It would be interesting to check if trajectories with initial angles of multiples of 30° tend to travel further from their starting positions than trajectories with other initial angles or if they simply fill in more of their local area. Another component of these graphs is that behavior is not only cyclic in 60° intervals, but there is a symmetry at 180° intervals. Each graph has a unique feature that occurs 180° apart, an example being the prominent gap in Figure 13c. Similar features may be seen in the three other graphs.

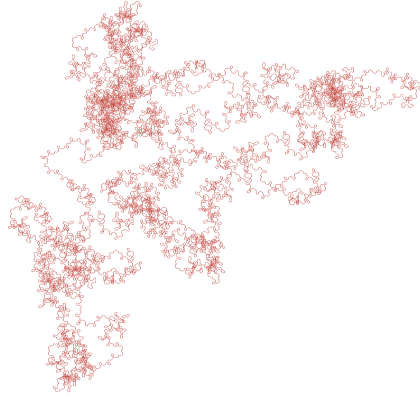


FIGURE 14. A 1000000-bounce trajectory with initial angle 90° . It appears to be periodic, though we have not verified this. A cycle was first detected at bounce 26016.

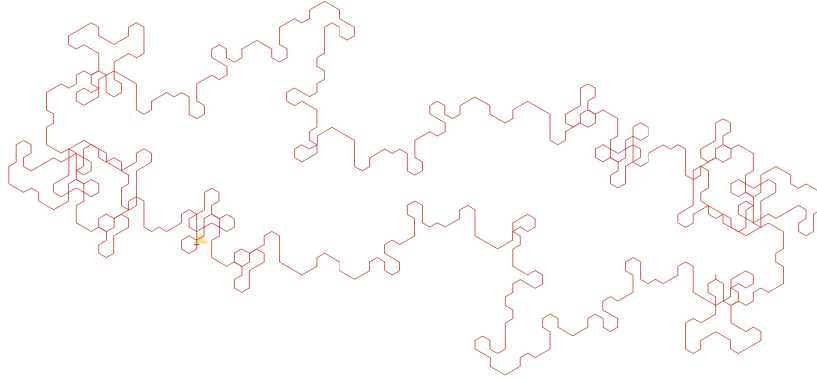


FIGURE 15. A 1000000-bounce trajectory with initial angle 90° . It appears to be periodic, though we have not verified this. Interestingly, this trajectory appears to be completely symmetric.

A striking feature of our simulations is that many trajectories appear to be periodic; this spans beyond the bucket described above. In most cases we have not verified periodicity exactly, so these trajectories may instead represent very long near-periodic orbits or numerical

artifacts; nevertheless, apparently closed behavior occurs frequently across the trajectories we have examined. A similar phenomenon was reported by Diana Davis in preliminary experiments on Penrose tiling billiards with Pat Hooper, where simulations suggested that almost every trajectory may be periodic [3]. The appearance of abundant periodic behavior in both settings suggests that periodicity may be a significant feature of tiling billiards on aperiodic tilings rather than an isolated property of the hat.

If the apparent abundance of periodic trajectories in both the Penrose and hat settings reflects a genuine dynamical phenomenon, it would be a remarkable feature of aperiodic tiling billiards. As a trajectory moves through an aperiodic tiling, its large-scale environment is not constrained to recur by translation as it would in a periodic tiling. These changing environments may provide repeated opportunities for trajectories to be redirected or reversed, potentially favoring return over indefinite escape. We suggest this phenomenon be explored further.

3.4. Path reversal. A prominent feature of many simulated trajectories is path reversal. After making an excursion, a trajectory may return along a route that closely follows its earlier path in the opposite direction. This behavior is distinct from the repeated local looping seen in patterns such as the Sigil. From our surveyed trajectories, this appears to happen most frequently in the thicket style, though it appears in a great deal of trajectories of all styles. In fact, this behavior is exhibited somewhere in nearly all of our simulated trajectories.

An observed mechanism for this behavior is when a trajectory runs around a vertex and re-enters the configuration of hats seen in the original excursion, only with the reverse direction. If a sufficiently wide range of nearby trajectories with the same angle follows the same local sequence of crossings and the displacement from the original path is sufficiently small upon path reversal, the trajectory may continue to follow its earlier route backwards for many bounces before separating from it. We show an example of this mechanism in Figure 16.

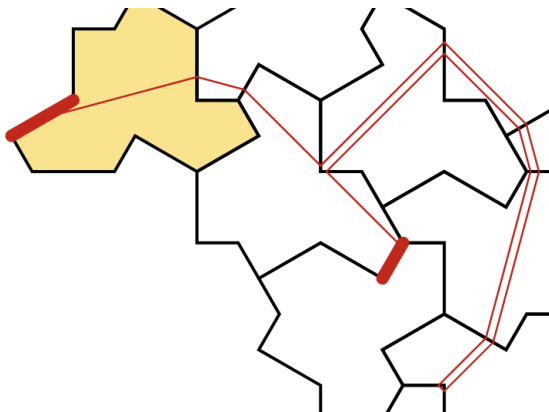


FIGURE 16. A trajectory segment that rounds a vertex and follows itself back temporarily.

To quantify large-scale displacement, we recorded the Euclidean distance between the trajectory's initial point and its position after n bounces. We refer to the resulting plot as a *bounce-distance graph*. The effects of path reversal are visible in such plots.

Bounce-distance graphs often contain substantial local and global symmetry simultaneously. Global symmetry can occur when a trajectory reverses a long excursion and returns near its initial region, while local symmetry can occur when it reverses only part of an earlier excursion before departing along a new route. The trajectories causing these graphs seem to make many

small excursions that are each reversed while the entire trajectory reverses as well. An example is shown in Figure 17.

Many trajectories seem to take long excursions but ultimately return to their starting side of their starting hat. The frequency of such return maps could be explored further.

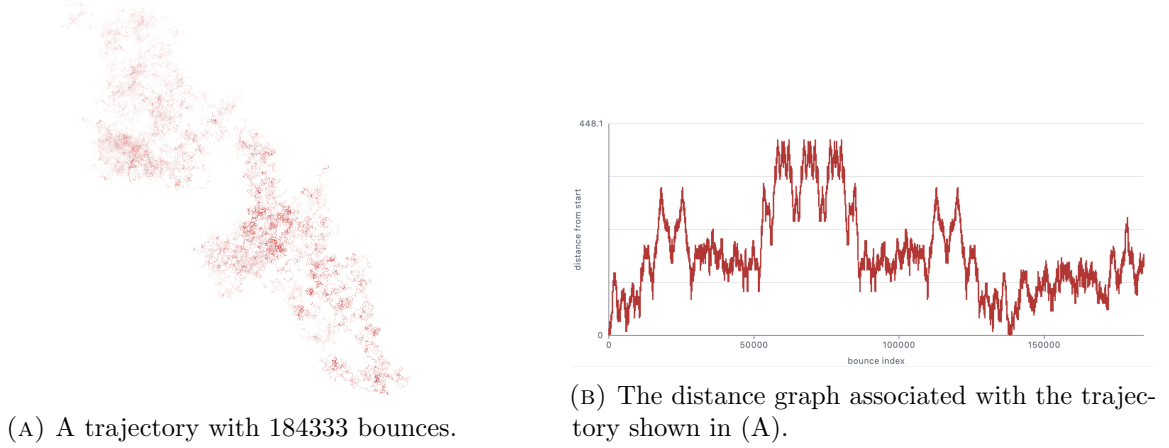


FIGURE 17. A bounce-distance graph exhibiting local and global symmetry up to a point. The trajectory returns extremely close to its initial position after following its global route backward.

3.5. Sensitivity to initial conditions. We also compared trajectories with nearby but distinct initial conditions. In the example shown in Figure 18, the trajectories have the same initial direction and begin on the same edge, but their initial positions differ slightly along that edge. They separate rapidly over time, providing computational evidence of finite-time sensitivity to initial position. Separate experiments in which the initial angle was perturbed showed similar divergence, suggesting finite-time sensitivity to both components of the initial condition. This was only tested on tens of trajectory starting conditions.

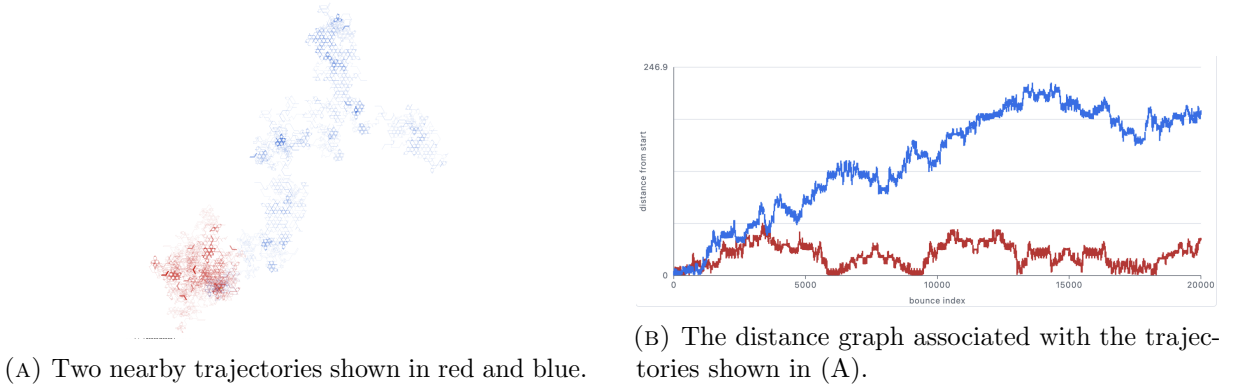


FIGURE 18. Two 20000-bounce trajectories with identical initial conditions aside from a displacement of 1% of their starting side's length. They diverge from each other rapidly, displaying sensitivity to initial conditions.

3.6. Additional trajectory structure. The apparent structure of these trajectories is not limited to periodic or locally retracing behavior. Long simulated trajectories also tend to exhibit

spatial clustering. A trajectory may spend a large number of bounces in one region, leave along a comparatively sparse bridge, and then enter another region of high activity. This behavior appears in all three observed trajectory styles.

We have not determined whether particular regions are intrinsically more active across many initial conditions or whether the apparent clustering is specific to each trajectory or due to the finite duration of the simulations. One possible direction for further study is to define an activity statistic for hat patches and compare it across supertiles of the same type and level. Such an analysis could reveal whether the observed clustering is related to the HTPF hierarchy.

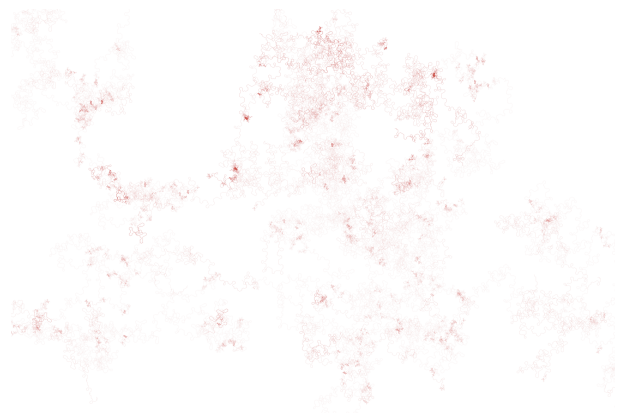


FIGURE 19. A long simulated trajectory exhibiting spatial clustering. Some regions are visited repeatedly, while comparatively sparse trajectory segments connect them.

The web structure of honeycomb-style trajectories suggests another possible form of organization. For one fixed honeycomb-style trajectory, we recorded successive occurrences of the Sigil and compared the sequences of hat crossings connecting one occurrence to the next. Some of these intervening sequences appeared more than once, so distinct visits to the Sigil can follow the same route through the tiling.

This experiment was limited to a single trajectory and does not by itself establish any substitution structure. Nevertheless, it raises the possibility that recurring trajectory patterns such as the Sigil could themselves serve as larger-scale building blocks for describing the dynamics. The apparent alignment of several recurring patterns with metatile structure suggests that this organization may also interact with the HTPF hierarchy.

These observations lead naturally to a quantitative question: how many distinct finite crossing patterns can occur in hat-tiling billiards? In the next section, we introduce a symbolic encoding of trajectories and use it to formulate this question precisely.

4. SYMBOLIC LANGUAGE COMPLEXITY

In this section we state and prove our theoretical result. We also present experimental evidence of sharper bounds than we were able to prove for hat tiling billiards language complexity. We discuss implications of these experiments, present conjectures, and suggest methods for their study. We also give thoughts on how these theories may apply to tilings by other members of the hat family.

A common method for studying billiard dynamics is to encode a trajectory by the sequence of edges it crosses. The symbols used in such an encoding (called *letters*) form an *alphabet*, and the resulting sequence is called a *word*. For trajectories on hat tilings, we use the following alphabet:

Definition 4.1 (Hat-trajectory alphabet). Let $\mathcal{S}$ denote the set of side labels of the hat and its reflection shown in Figure 2a. The *hat-trajectory alphabet* $\mathcal{A}$ is the set of all ordered pairs

$$a \rightarrow b,$$

with $a, b \in \mathcal{S}$, for which there exist two adjacent hats and a nonsingular trajectory that exits the first hat through the side labeled a and enters the second hat through the side labeled b . Thus, each symbol records both the exit side of the current hat and the entry side of the next hat. Only pairs realized by legal hat adjacencies are included in $\mathcal{A}$.

A singular trajectory has no uniquely defined continuation beyond its first vertex hit, so its symbolic coding terminates there.

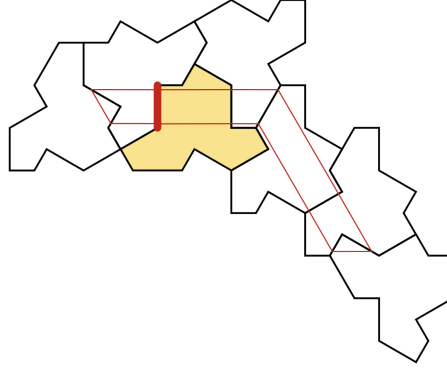


FIGURE 20. The first 12 bounces of the trajectory in Figure 1. Its associated symbolic word is $8 \rightarrow 0, 2 \rightarrow 2, 9 \rightarrow 4, 7 \rightarrow 1, 0 \rightarrow 8, 4 \rightarrow 9, 2 \rightarrow 2, 0 \rightarrow 8, 12 \rightarrow 4, 8 \rightarrow 12, 11 \rightarrow 7, 4 \rightarrow 12$. This segment forms the recurring pattern called the V, discussed in Section 3.

Because hat tilings are aperiodic, identical finite trajectory segments may occur in distinct large-scale geometric contexts and need not have the same continuation. It is therefore natural to ask how many distinct finite symbolic patterns can occur along a single trajectory. The language complexity function measures this quantity:

Definition 4.2 (Language complexity). Let w be a finite or infinite word over an alphabet $\mathcal{A}$. For each positive integer n , let $\mathcal{L}_n(w)$ denote the set of all distinct contiguous factors of length n occurring in w . The *language complexity function* of w is defined by

$$p_w(n) = |\mathcal{L}_n(w)|$$

for every integer $n \geq 1$.

As mentioned above, this definition applies only to individual trajectories. It can be extended to cover all possible trajectories simultaneously:

Definition 4.3 (Global language complexity). Let $\mathfrak{T}$ denote the collection of all Hat tilings in the sense of Definition 2.4; in particular, every $\tau \in \mathfrak{T}$ has a single global HTPF hierarchy. For $n \geq 1$, let

$$\mathcal{L}_n^{\text{glob}} = \{u \in \mathcal{A}^n : \exists \tau \in \mathfrak{T} \text{ and } \gamma \in \Gamma_\tau \text{ whose symbolic word contains } u \text{ as a factor} \}.$$

The *global language-complexity function* is

$$P_{\text{glob}}(n) = |\mathcal{L}_n^{\text{glob}}|.$$

We prove a polynomial bound on the global language complexity of tiling billiards on hat tilings, showing the system features topological entropy zero. We believe tighter bounds to be possible for the global language complexity and provide possible ranges. We also conjecture the complexity of individual trajectories to be linear and suggest methods for exploration.

Theorem 4.4 (Quintic global complexity). *There exists a constant $K > 0$ such that*

$$P_{\text{glob}}(n) \leq Kn^5$$

for every integer $n \geq 1$.

4.1. Strategy. The proof proceeds by matching symbolic and geometric scales. For a given word length n , we choose a hierarchy level whose characteristic supertile diameter is comparable to n , and use the finite, scale-independent combinatorics of the HTPF hierarchy to show that every length- n word occurs in one of uniformly finitely many coarse geometric environments of diameter $O(n)$. After fixing such an environment, we fold the relevant trajectories into straight lines and organize them by direction and perpendicular displacement. The relevant grid vertices determine critical curves in this line parameter space, whose complementary regions have fixed edge-crossing itineraries. There are $O(n^2)$ relevant vertices and hence $O(n^4)$ line regions, while each region admits only $O(n)$ starting positions along its common itinerary. Thus each coarse environment produces $O(n^5)$ words, and the uniform bound on the number of environments gives the same estimate globally. The next subsection establishes the comparison between word length and supertile diameter required for the initial reduction.

4.2. Comparative growth scales. Before this renormalized argument can be carried out, one must reconcile the symbolic length of a trajectory with the geometric scale of a supertile. Each symbol records a bounce from one hat to an adjacent hat and therefore counts one step in G_τ ; it does not record the Euclidean length of the corresponding trajectory segment. The HTPF hierarchy gives exponential growth in Euclidean size, but the proof requires the analogous growth statement for diameter in G_τ .

For a level- k supertile S_k , let P_{S_k} denote its associated hat patch. Define the graph diameter of S_k by

$$\text{diam}_G(S_k) = \max \{d_{G_\tau}(H, H') : H, H' \in P_{S_k}\},$$

where distance is measured in the ambient hat adjacency graph G_τ .

We first identify the exponential scale of the HTPF hierarchy. Let

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \Lambda = \varphi^2 = \frac{3 + \sqrt{5}}{2}.$$

The regular area growth factor of the HTPF supertiles is φ^4 ; see [9, Figure 2.11 and Section 2]. Since area is proportional to the square of linear scale, the corresponding asymptotic linear growth factor is

$$\sqrt{\varphi^4} = \varphi^2 = \Lambda.$$

The finite-level H , T , P , and F supertiles are not exact scaled copies of the corresponding supertiles at the preceding level. However, their Euclidean diameters are uniformly comparable to Λ^k . The following lemma shows that the same exponential scale governs their diameters in the hat adjacency graph.

Lemma 4.5 (Exponential growth of supertile diameter). *There exist constants $c, C > 0$ and an integer $k_0 \geq 1$ such that, for every $k \geq k_0$ and every level- k supertile S_k ,*

$$c\Lambda^k \leq \text{diam}_G(S_k) \leq C\Lambda^k.$$

Proof. Let

$$\text{diam}_E(S_k) = \max \{ \|x - y\| : x, y \in |P_{S_k}| \}$$

denote the Euclidean diameter of the geometric realization of P_{S_k} .

The asymptotic linear growth factor of the HTPF hierarchy is Λ , and the normalized H , T , P , and F supertile shapes converge to nondegenerate limiting shapes, as computed in [9, p. 17]. Consequently, there exist constants $c_E, C_E > 0$ such that

$$c_E \Lambda^k \leq \text{diam}_E(S_k) \leq C_E \Lambda^k$$

for every sufficiently large k and every level- k supertile S_k .

It remains to compare Euclidean diameter with graph diameter. Let D denote the Euclidean diameter of a single hat.

First, suppose that

$$H = H_0, H_1, \dots, H_m = H'$$

is a path in G_τ . Consecutive hats in this path share a line segment of positive length. Moving through the hats along these shared segments shows that the Euclidean distance between any point of H and any point of H' is at most

$$(m + 1)D.$$

Choose points $x, y \in |P_{S_k}|$ whose distance is $\text{diam}_E(S_k)$, and choose hats $H, H' \in P_{S_k}$ containing x and y , respectively. Applying the preceding estimate to a shortest path from H to H' gives

$$\text{diam}_E(S_k) \leq D(\text{diam}_G(S_k) + 1).$$

Therefore,

$$\text{diam}_G(S_k) \geq \frac{c_E}{D} \Lambda^k - 1.$$

By increasing k_0 , the constant term may be absorbed into the exponential term. Hence there is a constant $c > 0$ such that

$$\text{diam}_G(S_k) \geq c \Lambda^k$$

for every $k \geq k_0$.

For the reverse inequality, choose hats $H, H' \in P_{S_k}$. If $H = H'$, then $d_{G_\tau}(H, H') = 0$, so assume that $H \neq H'$.

The tiling contains only countably many hats. Indeed, each hat contains a point with rational coordinates in its interior, and distinct hats have disjoint interiors. Since each hat has finitely many vertices and sides, the tiling therefore has only countably many vertices and only countably many lines containing hat boundaries.

Choose an interior point $x \in H$ that lies on none of the lines containing a hat boundary. Such a choice is possible because a countable union of lines has area zero, whereas the interior of H has positive area.

With x fixed, each vertex v of the tiling determines a line passing through x and v . There are only countably many such lines. We may therefore choose an interior point $y \in H'$ that lies on none of them.

The straight segment joining x to y passes through no vertex of the tiling. It also cannot contain a positive-length portion of a hat boundary, since this would require x to lie on the line containing that boundary. Thus the segment remains a single straight line and crosses from one hat to the next only through boundary segments of positive length. The hats encountered along the segment consequently form a walk in G_τ .

Let

$$R = \|x - y\|.$$

Every hat met by the segment contains at least one point of the segment. Since a single hat has Euclidean diameter D , the entire hat lies within distance D of the segment. Thus all hats met by the segment are contained in its D -neighborhood.

The D -neighborhood of a segment of length R consists of a rectangle of length R and width $2D$, together with two semicircular ends of radius D . Its area is therefore

$$2DR + \pi D^2.$$

Let $A > 0$ denote the area of a single hat, and let N denote the number of distinct hats met by the segment. Since the hats have disjoint interiors,

$$N \leq \frac{2DR + \pi D^2}{A}.$$

Because the hat is not convex, the segment may meet the same hat more than once. Nevertheless, there is a uniform constant q such that the intersection of any line with a single hat hits that hat at most q times. Indeed, the hat is a fixed polygon with finitely many sides, so a line can cross its boundary only a uniformly bounded number of times.

It follows that the walk determined by the segment visits each distinct hat at most q times. Hence

$$d_{G_\tau}(H, H') \leq qN \leq \frac{2qD}{A}R + \frac{q\pi D^2}{A}.$$

Choose $C_0 > 0$ so that

$$C_0 \geq \max \left\{ \frac{2qD}{A}, \frac{q\pi D^2}{A} \right\}.$$

Then

$$d_{G_\tau}(H, H') \leq C_0 R + C_0 = C_0(R + 1).$$

Since $R = \|x - y\|$, we obtain

$$d_{G_\tau}(H, H') \leq C_0(\|x - y\| + 1).$$

The constant C_0 depends only on the fixed geometry of a single hat and is therefore independent of H , H' , and k .

The resulting path is allowed to leave P_{S_k} , since $\text{diam}_G(S_k)$ is defined using distance in the ambient graph G_τ . Because $x, y \in |P_{S_k}|$, we obtain

$$d_{G_\tau}(H, H') \leq C_0(\text{diam}_E(S_k) + 1).$$

Taking the maximum over all $H, H' \in P_{S_k}$ gives

$$\text{diam}_G(S_k) \leq C_0(\text{diam}_E(S_k) + 1).$$

Using the Euclidean upper bound and again increasing k_0 to absorb the constant term, we conclude that

$$\text{diam}_G(S_k) \leq C\Lambda^k$$

for some constant $C > 0$. □

The finitely many levels below k_0 do not affect the asymptotic complexity estimate. We therefore begin the hierarchical argument at level k_0 . For every $k \geq k_0$, define

$$L_k = \left\lceil \Lambda^k \right\rceil.$$

By Lemma 4.5, L_k is uniformly comparable to the graph diameter of every level- k supertile.

4.3. Bounded supertile environments. Fix $n \geq L_{k_0}$, and choose k so that

$$L_k \leq n < L_{k+1}.$$

Since $L_k \asymp \Lambda^k$ and the ratio between consecutive characteristic lengths is uniformly bounded, this choice gives

$$\Lambda^k \asymp n.$$

We first use this scale to reduce the global counting problem to finitely many coarse geometric environments.

Lemma 4.6 (Bounded coarse environments). *There exist integers $J, M \geq 1$ with the following property. Let $\mathcal{S}$ be the collection of level- k supertiles whose hat patches contain hats visited by an n -bounce trajectory segment. Then:*

- (1) $\mathcal{S}$ contains at most M level- k supertiles;
- (2) the union of their hat patches has Euclidean diameter $O(n)$ and contains $O(n^2)$ hats;
- (3) after absolute position is ignored, $\mathcal{S}$ belongs to one of at most J possible arrangements.

An arrangement records the type and orientation of each supertile and the boundary identifications between neighboring supertiles. The constants are independent of n , k , the tiling, and the trajectory.

Proof. Let

$$H_0, H_1, \dots, H_n$$

be the successive hats visited by the trajectory segment, and choose a point $x \in H_0$. Let D be the Euclidean diameter of one hat.

Consecutive hats share a boundary segment of positive length. It follows that every point of H_j lies within distance at most $(j+1)D$ of x . Hence all visited hats lie in a disk of radius $(n+1)D$.

Now let $S \in \mathcal{S}$. By the definition of $\mathcal{S}$, the patch P_S contains at least one hat visited by the trajectory segment. Choose one such hat and denote it by Q_S .

Let $y \in |P_S|$, and choose any point $z \in Q_S$. Since both y and z belong to the patch of S ,

$$\|y - z\| \leq \text{diam}_E(S).$$

The upper diameter estimate established in the preceding subsection is uniform over all level- k supertiles, so there exists $\Delta_k > 0$ such that $\text{diam}_E(S) \leq \Delta_k$ uniformly for every $S \in \mathcal{S}$. Therefore every point in the union of the patches P_S lies within distance Δ_k of some hat visited by the trajectory segment.

Let $B(x, R_n)$ be the disk containing all hats visited by the trajectory segment, where $R_n = (n+1)D$. Now take any $S \in \mathcal{S}$ and any point $y \in |P_S|$. By the definition of $\mathcal{S}$, the patch P_S contains a visited hat; choose one and denote it by Q_S . For any point $z \in Q_S$, both y and z belong to P_S , and hence

$$\|y - z\| \leq \text{diam}_E(S) \leq \Delta_k.$$

Since Q_S is a visited hat, $z \in B(x, R_n)$. The triangle inequality therefore gives

$$\|y - x\| \leq \|y - z\| + \|z - x\| \leq \Delta_k + R_n.$$

Because S and y were arbitrary, this proves the explicit containment

$$\bigcup_{S \in \mathcal{S}} |P_S| \subseteq B(x, R_n + \Delta_k).$$

The disk containing all the hats in

$$\bigcup_{S \in \mathcal{S}} |P_S|$$

has radius

$$R_n + \Delta_k = O(n),$$

since $\Delta_k \asymp L_k \asymp n$, so the union has Euclidean diameter $O(n)$ and is contained in a disk of area

$$\pi(R_n + \Delta_k)^2 = O(n^2).$$

Every level- k supertile has area

$$\Theta(\Lambda^{2k}) = \Theta(n^2),$$

uniformly over the four supertile types. Since distinct level- k supertile patches have disjoint interiors, only a bounded number can fit inside the $O(n^2)$ -area disk above. Thus there is a constant M , independent of n and k , such that

$$|\mathcal{S}| \leq M.$$

The same disk estimate bounds the number of hats in the union. Every hat has the same positive area A , and distinct hats have disjoint interiors. Therefore the union contains only $O(n^2)$ hats.

It remains to count the possible arrangements of the supertiles. The visited hats form a connected walk, so the level- k supertiles containing them form a connected collection. At every hierarchy level there are four supertile types, finitely many orientations, and finitely many permitted ways for neighboring supertiles to meet. These choice sets do not grow with k , because the HTPF decomposition at each level is combinatorially equivalent to the decomposition at the preceding level; see [9, Section 5].

Consequently, only finitely many arrangements are possible. Since M and all local choice sets are independent of k , their total number is bounded by a constant J independent of n , k , the tiling, and the trajectory. $\square$

Lemma 4.6 removes the large-scale aperiodicity from the counting problem. For the remainder of the argument, fix one collection $\mathcal{S}$ described by the lemma. For each $S \in \mathcal{S}$, let P_S denote its hat patch, and set

$$U_{\mathcal{S}} = \bigcup_{S \in \mathcal{S}} |P_S|.$$

Then

$$\text{diam}_{\mathbb{E}}(U_{\mathcal{S}}) = O(n).$$

4.4. Folding trajectories into a finite grid patch. A labeled side of a hat may consist of several consecutive edges of the underlying [3.4.6.4] Laves grid, and different portions of the same labeled side may face different neighboring hats. We therefore subdivide every labeled side at its intermediate grid vertices. Each resulting boundary piece is one grid edge and retains the label of the original hat side.

This subdivision does not change the symbolic word. Its purpose is only to ensure that crossing a boundary piece determines one neighboring hat.

Fold each trajectory segment back to a straight segment in the usual way, keeping its first hat fixed. Denote the folded copies of the successive hats by

$$\hat{H}_0, \hat{H}_1, \dots, \hat{H}_m, \quad m \leq n.$$

Consecutive folded hats share the boundary piece crossed by the trajectory.

Reflection across the line containing any edge of the [3.4.6.4] Laves grid preserves the grid. Consequently, every boundary edge of every folded hat belongs to one fixed copy of the Laves grid. The folded hats may overlap, but this does not affect the argument.

Recall D is the Euclidean diameter of one hat. For any $y \in \widehat{H}_j$, choose a point $z \in \widehat{H}_0$. For each $1 \leq i \leq j$, choose a point

$$q_i \in \widehat{H}_{i-1} \cap \widehat{H}_i$$

on the boundary piece shared by the two folded hats. The points z and q_1 lie in $\widehat{H}_0$, the points q_i and q_{i+1} lie in $\widehat{H}_i$, and q_j and y lie in $\widehat{H}_j$. Since every folded hat has diameter D ,

$$\|y - z\| \leq \|z - q_1\| + \sum_{i=1}^{j-1} \|q_{i+1} - q_i\| + \|q_j - y\| \leq (j+1)D.$$

For $j = 0$, the same inequality follows directly from the diameter of $\widehat{H}_0$.

This estimate places each folded chain within $O(n)$ of its own first hat. We now place the folded chains arising from all possible first hats inside one common disk. By the preceding lemma, there is a constant $C_0 > 0$ such that

$$\text{diam}_E(U_S) \leq C_0 n.$$

Fix $x_0 \in U_S$. Since the first hat is kept fixed,

$$\widehat{H}_0 \subseteq U_S,$$

so the point z chosen above satisfies

$$\|z - x_0\| \leq C_0 n.$$

It follows that

$$\|y - x_0\| \leq \|y - z\| + \|z - x_0\| \leq (j+1)D + C_0 n \leq (n+1)D + C_0 n.$$

Therefore every folded hat arising from every trajectory under consideration is contained in the single disk

$$B_n = \overline{B}(x_0, C_0 n + (n+1)D).$$

In particular, B_n has radius $O(n)$.

Let E be the set of Laves-grid edges that meet B_n :

$$E = \{e : e \text{ is a Laves-grid edge and } e \cap B_n \neq \emptyset\}.$$

Every boundary edge of every folded hat belongs to E .

Let r_* be the maximum length of a Laves-grid edge, and enlarge B_n to

$$B_n^+ = \{x \in \mathbb{R}^2 : \text{dist}(x, B_n) \leq r_*\}.$$

If an edge meets B_n , each of its endpoints lies within distance r_* of a point of B_n . Hence both endpoints of every edge in E lie in B_n^+ .

Let V be the set of Laves-grid vertices contained in B_n^+ , and write

$$N = |V|.$$

The Laves grid is built from a single kite with uniform area, so the number of its vertices in a disk is bounded by a constant multiple of the disk's area. Since B_n^+ has radius $O(n)$, it follows that

$$N = O(n^2).$$

Throughout this construction, the first hat remains in its original position in the fixed arrangement U_S . We do not translate or rotate the folded configuration after straightening it.

4.5. The line diagram and edge-crossing itineraries. We now organize the possible oriented folded lines. Fix an origin in the plane. For $0 \leq \theta < 2\pi$, define

$$u_\theta = (\cos \theta, \sin \theta), \quad \nu_\theta = (-\sin \theta, \cos \theta).$$

For $p \in \mathbb{R}$, let

$$\ell_{\theta,p} = \{z \in \mathbb{R}^2 : z \cdot \nu_\theta = p\},$$

oriented in the direction u_θ . Thus θ specifies the direction of travel and p specifies the signed perpendicular displacement from the origin.

We represent these lines by the parameter space

$$0 \leq \theta < 2\pi, \quad p \in \mathbb{R},$$

with the boundaries $\theta = 0$ and $\theta = 2\pi$ identified. We call this parameter space the *line diagram*.

For a vertex $v = (v_1, v_2) \in V$, the line $\ell_{\theta,p}$ passes through v exactly when

$$p = v \cdot \nu_\theta = -v_1 \sin \theta + v_2 \cos \theta.$$

The parameters representing oriented lines through v therefore form the curve

$$C_v = \{(\theta, p) : p = -v_1 \sin \theta + v_2 \cos \theta\}$$

in the line diagram.

Remove all the curves C_v , for $v \in V$, and call each path-connected component of the complement a *line region*. Thus, within one line region, the corresponding physical line can move continuously without passing through a vertex of V .

For a parameter (θ, p) in a line region, follow the oriented line $\ell_{\theta,p}$ in its direction of travel and list the edges of E that it crosses:

$$\mathcal{I}_E(\theta, p) = (e_1, e_2, \dots, e_m).$$

We call this ordered list the E -itinerary of the line.

Lemma 4.7 (Constancy of line-region itineraries). *The E -itinerary is constant throughout each line region. Moreover, every length- n word produced inside U_S is represented by a folded line lying in some line region.*

Proof. Let (θ, p) move continuously inside one line region. We first show that the set of edges crossed by the corresponding physical line cannot change.

Fix a line region $\mathcal{R}$ and two oriented lines ℓ_0, ℓ_1 represented by points of $\mathcal{R}$. Since $\mathcal{R}$ is path-connected, there is a continuous deformation from ℓ_0 to ℓ_1 whose parameter remains inside $\mathcal{R}$.

Now fix an arbitrary edge $e \in E$. If ℓ_0 crossed e and ℓ_1 did not, then at some intermediate point of the deformation the intersection with e would have to reach an endpoint v of e . The intermediate line would then pass through v , so its parameter would lie on C_v . This is impossible because C_v is disjoint from $\mathcal{R}$. Thus ℓ_0 crosses e if and only if ℓ_1 crosses e .

Since this holds for every $e \in E$, all lines represented in $\mathcal{R}$ cross exactly the same set of edges.

It remains to show that these edges are crossed in the same order. Fix two distinct edges $e, f \in E$ that are crossed by the lines in $\mathcal{R}$, and let ℓ_t , for $0 \leq t \leq 1$, be a continuous deformation between two such lines. Denote the intersections of ℓ_t with e and f by $x_e(t)$ and $x_f(t)$, respectively. These intersection points move continuously as the line moves.

Suppose that, when traveling along the oriented line ℓ_0 , one encounters $x_e(0)$ before $x_f(0)$, but along ℓ_1 one encounters $x_f(1)$ before $x_e(1)$. For their order to reverse, there must be some intermediate value t_* at which

$$x_e(t_*) = x_f(t_*).$$

This common point belongs to both e and f . Distinct edges of the Laves grid can meet only at a grid vertex, so ℓ_{t_*} passes through some vertex $v \in V$. Its parameter therefore lies on C_v , contradicting the fact that the deformation remains inside $\mathcal{R}$.

Thus the relative order of every pair of crossed edges is fixed, and hence the complete E -itinerary is constant throughout $\mathcal{R}$.

Since this argument applies to every edge in E and every pair of crossed edges, the complete ordered list

$$\mathcal{I}_E(\theta, p)$$

is constant throughout each line region.

The line regions contain only parameters that lie on none of the vertex curves C_v . We must therefore show that restricting attention to these parameters does not omit a word whose original folded line has parameter (θ_0, p_0) on one or more vertex curves not associated with a chosen length- n word.

Fix such a word. Because the realizing trajectory is nonsingular, each of its recorded crossings lies strictly inside a boundary edge. There is therefore an open neighborhood $\mathcal{U}$ of (θ_0, p_0) such that every parameter in $\mathcal{U}$, using the same starting hat and entering edge, produces the same first n boundary crossings and hence the same word.

Each vertex curve C_v has empty interior, and there are only finitely many vertices in V . Their union cannot cover the open set $\mathcal{U}$. We may therefore choose

$$(\theta', p') \in \mathcal{U} \setminus \bigcup_{v \in V} C_v.$$

This parameter belongs to a line region, and its corresponding folded line produces the original length- n word. Hence restricting the count to line regions does not omit any word. $\square$

4.6. Counting the line regions.

Lemma 4.8 (Number of line regions). *The line diagram contains $O(n^4)$ line regions.*

Proof. Consider two distinct vertices

$$v = (v_1, v_2), \quad w = (w_1, w_2)$$

in V . An intersection of C_v and C_w must satisfy

$$-v_1 \sin \theta + v_2 \cos \theta = -w_1 \sin \theta + w_2 \cos \theta.$$

Equivalently,

$$(w_1 - v_1) \sin \theta + (v_2 - w_2) \cos \theta = 0.$$

Since $v \neq w$, the two coefficients are not both zero. We may therefore write the left-hand side as

$$R \sin(\theta + \phi)$$

for some $R > 0$ and some angle ϕ . Over $0 \leq \theta < 2\pi$, the equation

$$R \sin(\theta + \phi) = 0$$

has exactly two solutions. Thus C_v and C_w can intersect at no more than two points.

There are $N = |V|$ vertex curves. Insert them into the line diagram one at a time. The j -th curve has at most $2(j-1)$ intersection points with the curves already present. These points divide the vertex curves into $O(j)$ arcs, each of which can create at most one new region. Hence the j -th curve creates at most $O(j)$ regions.

Summing over all N curves gives

$$O\left(\sum_{j=1}^N j\right) = O(N^2).$$

Since $N = O(n^2)$, the number of line regions is

$$O(N^2) = O(n^4).$$

□

4.7. From line regions to symbolic words. A line region determines the complete E -itinerary of a folded line, but it does not specify where the length- n trajectory segment begins. We therefore supplement a line region with a *starting choice*, consisting of a hat $H \subset U_{\mathcal{S}}$ and a subdivided boundary edge

$$e_{\text{in}} \subset \partial H$$

through which the oriented line enters H . Once the line and e_{in} are fixed, the starting point is their intersection.

Lemma 4.9 (Determinacy from a marked line region). *Fix a line region $\mathcal{R}$, a starting hat $H \subset U_{\mathcal{S}}$, and a subdivided entering edge $e_{\text{in}} \subset \partial H$. Among lines in $\mathcal{R}$ that cross e_{in} into H , these data determine at most one length- n word.*

Proof. By Lemma 4.7, every line in $\mathcal{R}$ has the same E -itinerary. After crossing e_{in} , every such line therefore encounters the boundary edges of the folded copy of H in the same order. In particular, the first boundary edge encountered after e_{in} is the same, so the exit edge of H is fixed.

Because the hat sides were subdivided into boundary edges with fixed neighboring hats, this exit edge determines the same next physical hat. Folding across the exit edge then determines the same next folded hat and its entering edge. Repeating this argument determines the exit edge at every subsequent crossing.

Thus the first n ordered pairs of labeled sides are identical. Some choices may leave $U_{\mathcal{S}}$ before completing n crossings and therefore produce no word, but the fixed data

$$(\mathcal{R}, H, e_{\text{in}})$$

cannot produce two different length- n words. □

Lemma 4.10 (Words in one coarse environment). *The number of length- n words produced by trajectory segments whose visited hats all lie in $U_{\mathcal{S}}$ is $O(n^5)$, uniformly over $\mathcal{S}$, k , and n .*

Proof. Fix a line region $\mathcal{R}$, and write its common E -itinerary as

$$\mathcal{I}_E(\mathcal{R}) = (e_1, \dots, e_m).$$

This is a one-dimensional crossing sequence through an $O(n)$ -scale patch of the fixed kite grid, so $m = O(n)$. A starting choice consists of selecting one of these m itinerary positions together with one of the finitely many possible local hat configurations there. Hence $\mathcal{R}$ admits $O(n)$ starting choices. □

Proof of Theorem 4.4. Let $n \geq L_{k_0}$, and choose k so that

$$L_k \leq n < L_{k+1}.$$

Every word

$$u \in \mathcal{L}_n^{\text{glob}}$$

is realized by an n -bounce trajectory segment in some hierarchical hat tiling. By Lemma 4.6, the level- k supertiles met by this segment belong to one of at most J coarse arrangements. By Lemma 4.10, each fixed arrangement produces at most

$$Cn^5$$

words, where C is independent of the arrangement, k , and n . It follows that

$$P_{\text{glob}}(n) \leq J C n^5$$

for every $n \geq L_{k_0}$.

Only finitely many positive integers satisfy $n < L_{k_0}$, and for each of them

$$P_{\text{glob}}(n) \leq |\mathcal{A}|^n < \infty.$$

Increasing the constant to absorb these finitely many cases gives $K > 0$ such that

$$P_{\text{glob}}(n) \leq K n^5$$

for every integer $n \geq 1$. □

Corollary 4.11. *The global symbolic system has topological entropy zero.*

Proof. Using the language-complexity definition of entropy,

$$h_{\text{top}} = \limsup_{n \rightarrow \infty} \frac{1}{n} \log P_{\text{glob}}(n).$$

Theorem 4.4 gives

$$0 \leq h_{\text{top}} \leq \limsup_{n \rightarrow \infty} \frac{\log K + 5 \log n}{n} = 0.$$

Therefore $h_{\text{top}} = 0$. □

The proof of Theorem 4.4 uses little of the geometry specific to the standard hat. Its essential components are the underlying kite grid, which makes the unfolding and line-counting argument possible, and the HTPF hierarchy, which supplies finitely many supertile types and controls their growth. Since these structures are shared by the other members of the hat family, we expect the same polynomial bound to hold throughout that family.

Conjecture 4.12 (Quintic global complexity for the hat family). Let T be any tile in the hat family, and let $P_{\text{glob}}^T(n)$ denote the global language complexity of nonsingular tiling billiard trajectories over all HTPF tilings by T , using the analogue of the alphabet in Definition 4.1. Then there exists a constant $K_T > 0$ such that

$$P_{\text{glob}}^T(n) \leq K_T n^5$$

for every integer $n \geq 1$.

Global language complexity is difficult to measure experimentally due to trouble with finding an accurate sample of trajectories of sufficient length, but our surveyed trajectories point towards a substantially tighter bound being possible. We chose 2000 random initial conditions in our level-6 H supertile and ran a 35000-bounce trajectory at each. We measured the global language complexity of this set for word length $1 \leq n \leq 20$. Given that new words of higher length are more likely to be discovered as trajectory length increases, we fit curves to our data along the full set of 20 word lengths and smaller sub-ranges. The data is displayed in Table 3 and we conjecture the true growth rate of global language complexity in Conjecture 4.13 based on this data.

TABLE 3. Estimated global complexity for $1 \leq n \leq 20$.

Word length n	Complexity
1	57
2	890
3	2045
4	3980
5	6323
6	9755
7	13648
8	18579
9	24093
10	30824
11	38261
12	46972
13	56454
14	67703
15	79997
16	93935
17	108935
18	125933
19	144137
20	164452

Conjecture 4.13 (Sharper global language complexity). Let $\alpha \approx 2.3$ – 2.4 . There exists $K > 0$ such that

$$P_{\text{glob}}(n) \leq Kn^\alpha$$

for all sufficiently large n .

Motivated by our experimentally observed growth exponents, we looked for naturally occurring exponents related to the geometry of the hat family. Baake, Gähler, and Sadun show that the Koch-type fractal portions of the acceptance-window boundaries associated with the CAP tiling have Hausdorff dimension

$$d_H = \frac{\log(2 + \sqrt{3})}{2 \log \phi} \approx 1.36837649$$

[1, Eq. (17)]. Consequently,

$$1 + d_H \approx 2.36837649,$$

which is close to several of our finite-range estimates for the global language-complexity exponent. At present, however, we know of no mechanism relating this Hausdorff dimension to tiling-billiard language complexity, and the numerical agreement may be coincidental.

It would be worth exploring whether similar experimental results are available on tilings by other members of the hat family.

Theorem 4.4 deliberately permits the tiling, the trajectory, and the starting position to vary with the word. It is therefore a bound for the entire global language, not a prediction of the

complexity seen along one fixed trajectory. A single trajectory explores only one tiling from one initial condition, and we expect its collection of factors to be much smaller. Our experiments lead to the following conjecture.

Conjecture 4.14. Fix a hat tiling τ and a nonsingular trajectory $\gamma \in \Gamma_\tau$. Let w_γ be its symbolic word under the alphabet of Definition 4.1. Then there exist constants $C_\gamma, D_\gamma \geq 0$ such that

$$p_{w_\gamma}(n) \leq C_\gamma n + D_\gamma$$

for every integer $n \geq 1$.

We explored two possible approaches to Conjecture 4.14.

Budgeted extension. Choose characteristic word lengths L_k comparable to the diameter of a level- k supertile, and let

$$B_k = L_{k+1} - L_k$$

be the number of additional symbols available when passing from one scale to the next. Rather than estimate complexity independently at every length, this method begins with the length- L_k factors of w_γ and follows their possible extensions through the next B_k symbols. The hierarchy is used to group occurrences that initially have the same local supertile environment and to identify the points at which their continuations become distinct. Each such split is then charged to the extension budget B_k . An estimate of $O(B_k)$ new branches between consecutive scales would give

$$p_{w_\gamma}(L_{k+1}) \leq p_{w_\gamma}(L_k) + O(B_k).$$

Since the budgets telescope, this would yield $p_{w_\gamma}(L_k) = O(L_k)$. The intermediate word lengths could then be handled using monotonicity and the bounded ratio between consecutive characteristic lengths.

S-adic representations and trajectory recognizability. We first extended the hat-crossing alphabet to an alphabet of metatiling crossings. The sides of the metatiles H , T , P , and F are the matching-rule sides used in the HTPF construction, so a metatiling symbol records the ordered pair consisting of the side through which the trajectory exits one metatiling and the side through which it enters the next. This is the direct analogue of the hat alphabet in Definition 4.1. The same construction can be repeated at every level of the supertiling hierarchy.

We then tried to construct a substitution system on these trajectory words. A level- k crossing symbol would be replaced by the block of level- $(k-1)$ crossings made beneath it. To make this replacement deterministic, we tried to enrich each symbol with a finite amount of memory recording its surrounding higher-level crossings. In this setting, trajectory recognizability means that, for every k , there is a finite memory length M_k such that a block of M_k consecutive level- k crossings uniquely determines the underlying level- $(k-1)$ word and its alignment. Such a family of deterministic refinements would give an S -adic representation of w_γ , allowing its complexity to be studied through the resulting finite substitution data.

5. CONCLUSION

Hat tiling billiards exhibit substantially more dynamical structure than might be expected from a system evolving on an aperiodic tiling. Our computational experiments and symbolic results approach this structure from different directions and together provide an initial picture of how billiard dynamics behave when translational periodicity is replaced by hierarchical organization.

Computationally, trajectories display persistent geometric regularity. We observed three broad visual styles determined only by initial direction, together with recurring local motifs, spatial clustering, extensive retracing, and path reversal. Most interestingly, ostensibly periodic behavior was abundant throughout our simulations. Although only the Sigil was verified exactly, many more complicated trajectories appeared to close, and we conjecture that every nonsingular

trajectory in the class of hat-edge directions is periodic, though some periods may be surprisingly long. Our experiments also reveal a strong directional dependence in trajectory exploration, including apparently opposite behavior between two natural classes of hat-edge directions for which we currently know no distinguishing dynamical mechanism. The prevalence of apparent periodicity is especially notable in light of Davis’ and Hooper’s preliminary experiments on Penrose tiling billiards, which similarly suggested that almost every trajectory may be periodic [3].

The symbolic dynamics provide a complementary form of structure. We prove that the global language complexity of hat tiling billiards grows at most quintically, and consequently that the associated global symbolic system has topological entropy zero. We expect this quintic bound to be far from sharp and conjecture the true global complexity to have exponent approximately 2.3–2.4. We also conjecture that every individual nonsingular trajectory has linear language complexity. Because the principal components of the global argument are shared throughout the hat family, we also expect analogous polynomial bounds for its other members.

Several central questions remain open. Foremost is the prevalence of periodic trajectories and the mechanism, if any, responsible for their apparent abundance. The conjectured periodicity of one class of edge directions gives a concrete special case, while the sharply different exploration behavior of the other natural edge-direction class remains unexplained. On the symbolic side, determining the true growth rate of the global language complexity and proving or disproving linear complexity for individual trajectories are the main problems left by our results. It is also natural to ask which of these phenomena persist throughout the hat family and which extend to other aperiodic tilings or even other substitution tilings.

The main conclusion of this study is therefore not merely that hat tiling billiards possess several unusual individual phenomena, but that removing translational periodicity does not appear to remove strong dynamical organization. The combination of hierarchical structure in trajectories, constrained symbolic complexity, and indications of widespread periodicity in both the hat and Penrose settings suggests that tiling billiards on aperiodic tilings may support a broader and unexpectedly rigid dynamical theory.

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